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Google Stadia: Game On or Game Over?

Introduction

Dov Zimring, Director of Product Management at Google, logged off his computer and stared out into the California sunset from his garage-turned-home-office. In one week, on August 6th 2020, Google's senior leadership, including CEO Sundar Pichai and CFO Ruth Porat, would be expecting an updated strategy and 5-year plan for Stadia, Google's cloud gaming service. Since its launch in November 2019, Stadia had struggled to gain adoption and user growth had been far below forecasts.

Zimring had founded Stadia inside of Google in 2013 with the vision of bringing high-end video games into the streaming age, much as Netflix had done for movies and Spotify for music. Games on Stadia would run on a server inside a Google data center instead of on a console in a user's living room, giving users the freedom to play anywhere, on any screen, without time-consuming downloads or expensive hardware. Achieving this vision would require multiple technological breakthroughs, but with the world's best engineers on staff, Google was no stranger to hard technical challenges. Zimring and his team had quietly developed and perfected the technology over the following six years. In 2018, Stadia released a preview of the product to an invite-only group of beta testers. The preview, codenamed *Project Stream*, featured a single game and set the gaming world abuzz. It was the first time a cloud gaming service had been able to deliver an experience comparable to playing on a traditional console.

Stadia was launched to the general public in November 2019. As with traditional gaming consoles, Stadia users still had to purchase individual games (in this case from Stadia's online store). But after that, all users needed to play their game was a screen and a stable internet connection - no consoles, extra storage drives, or downloads were required. Users could optionally pay $9.99 per month to upgrade to Stadia Pro, a high-end offering with premium features such as streaming games in 4K resolution and access to Google's most powerful gaming servers. Users who opted-out had access to Stadia Base, which was free but limited users to streaming in 1080p resolution[a].

Despite the momentum from *Project Stream*, Stadia was met with lackluster reception. Stadia's business model had been built around Stadia Pro as its main revenue stream, and most of the

[a] "4K resolution" refers to the number of pixels. More pixels resulted in a higher quality image and gameplay experience. 4K resolution had about 8M pixels and 1080p resolution (also called "HD" resolution) had about 2M pixels.

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leadership team was focused on growing the number of Pro subscribers. But even six months in, sign-ups were far short of Google's forecasts. Moreover, soon after Stadia's launch, both Microsoft and Sony announced their next generation consoles: the Xbox Series X and PlayStation 5.

Zimring needed to present a plan to Pichai and Porat outlining how Stadia was going to grow its user base. Some of Zimring's colleagues believed that Stadia should continue to vie for the "core" gamer segment through Stadia Pro. After all, this segment drove the most spend and were influencers for the rest of the market. If Stadia continued to invest in building the highest performance gaming servers and acquiring the latest hit games, these users might come to see the benefits of cloud gaming and switch over from traditional consoles. Others believed that Stadia should refocus its efforts on the "casual" gamer segment, who gravitated toward Stadia Base. But it wasn't clear how to monetize users who weren't willing to pay for games (which often sold for $60 or more) or premium features.

Either way, Zimring needed to decide on a course of action soon. Without meaningful revenue growth, the enormous cost of running Stadia's data centers would overwhelm the team's budget. And with Covid-19 putting a strain on the firm's advertising business, Google leadership was not keen to continue funding underperforming divisions.

The Video Games Industry

Video games had become a global entertainment phenomenon. In 2020, the industry brought in revenues of over $180 billion, eclipsing the global music and movie industries, which had revenues of $19 billion and $42 billion respectively.[1,2] The video games industry had two major segments: (1) mobile gaming and (2) console & PC gaming (see **Exhibit 1**).[3] Within each segment was a vibrant ecosystem of game platforms, game studios, and customers, also called "gamers."

Game platforms

The first video game was introduced in the 1960s when researchers at the Massachusetts Institute of Technology invented *Spacewar!*, a space combat video game for the PDP-1, an early mainframe computer.[4] While this primitive game captured the imagination of hobbyists, it remained inaccessible to most people since computers were not yet available to the layperson. This changed with the introduction of the Magnavox Odyssey, the first video game "console." Consoles were special computers designed to run video games and that could be connected to a living room TV set. Over the next two decades, multiple firms vied for the lucrative console market, producing hits like the Atari VCS, Sega Genesis, and Nintendo Entertainment System. In 1994, Sony introduced the PlayStation, one of the first consoles with 3D graphics capabilities. The PlayStation was an absolute sensation; its successor, the PlayStation 2, went on to become the best selling console of all time with over 155 million units sold.[5] In 2001, Microsoft also entered the market with the Xbox.

Up to this point, consoles and PCs were the dominant gaming platforms. But in 2007, Steve Jobs introduced the first smartphone to the world. The iPhone put a miniature gaming platform in everyone's pocket and thus mobile gaming was born. Apple's iOS and Google's Android mobile operating systems, with over 99% market share, became the dominant mobile gaming platforms. However, smartphones' small form factor, limited storage, and power constraints posed challenges for traditional console games such as *Call of Duty*, *FIFA,* and *Grand Theft Auto*. These games (often referred to as "AAA" titles) routinely required over 75 gigabytes of storage and had graphics that pushed the boundary of technology in each generation. On the other hand, mobile games, such as the popular *Angry Birds* or *Candy Crush*, had simpler graphics and gameplay and had a much smaller footprint. As a result, mobile and console games developed as largely separate markets.

Console and mobile platforms had different business models as well. Consoles earned revenue through hardware and game sales. When a consumer purchased an Xbox game from a retailer, the retailer and Microsoft would earn a margin and the remainder would go to the developer. Mobile platforms, on the other hand, charged a 30% fee on transactions facilitated on the platform, such as game purchases, in-app purchases, and more. Mobile platforms also facilitated the placement of ads in free games. In 2023, three firms dominated the console market: Sony with 45% of the market, and Microsoft and Nintendo with 27% each.[6] Apple and Google dominated mobile gaming, jointly accounting for over 99% of the market.

Game studios

Game studios were responsible for the actual design, development, and in the case of the largest studios, the publication and marketing of games. Creating a hit game took 4-5 years and a team with a rare blend of programming, design, art, and production skills. Moreover, even hit games only had a shelf life of 2-3 years, after which sales would decline in favor of newer games. Due to these dynamics, only a handful of studios were consistently profitable, leading to widespread consolidation that resulted in a few leading "AAA" game studios, such as Electronic Arts and Activision-Blizzard, followed by a long tail of "indie" studios. In recent years, platform makers Microsoft and Sony had been buying up studios; notable deals included Microsoft's acquisition of Activision Blizzard for $75 billion in 2023 and Sony's purchase of Bungie in 2022.[7]

The traditional business model for game studios was to sell games through third-party physical and digital retail channels, such as GameStop or Microsoft's online store. Succeeding in retail required large marketing and promotional budgets, which only AAA studios had. AAA games would often retail for $60 or more. While the traditional "buy-to-play" (B2P) model dominated, especially in AAA console games, the mobile gaming revolution had given rise to a new "free-to-play" (F2P) model. The advent of smartphones and app stores had driven the distribution cost of mobile games to near zero. This allowed studios to make the games themselves free and monetize through optional in-app purchases such as power-ups, additional levels, character skins, and more. Because only a tiny fraction of users made in-app purchases, F2P games needed to have a large user base. Thus, studios designed F2P games to run on as many platforms as possible, often bridging consoles and smartphones. For example, the hugely popular *Fortnite* was free to play and available on most devices, from PCs to consoles to smartphones. **Exhibit 2** shows the recommended GPU[b] for running *Fortnite* versus *CyberPunk 2077*, one of the top F2P games of 2020. Given the success of the F2P model, many studios had begun to opt for it over the traditional model.

Gamers

The term "gamer'' encompassed hundreds of millions of players around the world who enjoyed video games in any medium, and to any extent. Gamers had a wide variety of preferences, interests, and tastes that informed their choice of games and game platforms. **Exhibit 3** illustrates one useful customer segmentation scheme. Some segments, such as the "Ultimate Gamer,'' were notable for their high willingness to spend on gaming (see **Exhibit 4**). These premium users were also opinion leaders in the industry, and other segments often looked to them for information on which platforms and games to buy. As a result, they were highly sought-after influencers by platforms and studios alike. Other more casual gamer segments included the "Mainstream Gamers," who were more cost-conscious

[b] A GPU, or Graphics Processing Unit, was a computer processor that specialized in graphics and video. Most high end video games required the use of a GPU.

and driven by value, as well as "Lapsed Gamers" and "Time Fillers," who only played games occasionally.

Google Stadia

Google

Founded in 1998 by Stanford PhD dropouts Larry Page and Sergey Brin, Google began as an Internet search engine based on the founders' novel PageRank algorithm. Google's search results were far superior to alternatives at the time. Unlike other search engines, Google was also free to use for individuals and businesses alike, a counterintuitive decision at the time. Instead, Google's revenue came from advertisements displayed on its search results pages. Google quickly exploded in popularity and by 2004 was the most visited site on the Internet.[8]

Following its outstanding success in search, Google developed several popular internet services, including Gmail, Google Maps, and Google Docs; and acquired others, such as YouTube and Android. Consistent with its roots, Google gave these services away for free and monetized through ads, though some products would eventually offer an optional paid tier, such as YouTube Premium. By 2020, Google had nine products with over one billion users each, and several others with hundreds of millions of users.

Founding Stadia

In 2010, high-speed Internet connectivity across the US was low, with fewer than 40% of people having a connection higher than 10 megabits per second (Mbps), roughly the amount needed to reliably stream a 1080p video (see **Exhibit 5**). As part of building Search and other products, Google had invested in a sprawling network of data centers connected with high bandwidth fiber optic cables. A few people at Google started imagining the possibilities that this fiber optic infrastructure created to give people a lightning fast connection of over 1 gigabit per second (1000 Mbps), enabling them to stream high-definition video and download large files instantly. Of course, given its portfolio of successful Internet products, Google stood to benefit handsomely from providing users with better Internet access, and thus the Google Fiber team was born.

Zimring joined Google Fiber as a founding team member in 2010. Zimring, a patent-holding software engineer who had earned an MBA from UCLA, was taken with the question: What new possibilities would greatly increased connectivity in the coming years create? During his time at Google, he had witnessed the explosive success of mobile gaming thanks to Android. By reviewing data collected by YouTube, he also knew that gaming was one of the fastest-growing entertainment categories. It was already a $100 billion market and on its way to $200 billion in the next decade.

Also in 2010, as Zimring was becoming more interested in the potential for cloud gaming, a start-up called OnLive announced the OnLive Game System, the first cloud-gaming product. While the OnLive offering received some favorable press, it was clear that the system was ahead of its time. Streaming quality on OnLive was poor and users would frequently suffer from high latency[c]. Zimring knew that Google, with its expertise in data centers and proprietary streaming algorithms, could

[c] Latency, or colloquially "lag," was the delay between entering an input into the game controller and seeing the result of that action on the screen. High latency would detract from the gaming experience.

deliver a much better product. By 2013, he had convinced Google's senior leadership to go after the cloud-gaming market and had formed a small team to begin prototyping his vision.

Stadia's Value Propositions

Since Google already had an established presence in the mobile gaming market with Android, Stadia's focus fell to the console and PC gaming market. The Stadia team believed that cloud gaming could offer several distinct advantages over traditional consoles.

Cost and Convenience. With Stadia, users no longer needed to purchase expensive consoles or PCs every few years to be able to play the latest AAA games. Furthermore, Stadia was much more convenient, in that games could be played anywhere, at home or on the go, and on any screen including an old laptop, a tablet, or a TV. Games on Stadia were also available on demand. With traditional consoles, users would have to download a digital copy of a game to their console, which took several hours, or they would have to buy a physical copy and wait for it to arrive in the mail.

High Performance. Stadia strived to give users a gaming experience that rivaled the best consoles and PCs at a lower price point. This meant Stadia servers would have the latest GPUs and deliver games at the best resolution possible. In 2019, this meant streaming games in 4K resolution at 60 frames per second. But in the future, Stadia envisioned streaming games at 8K resolution and 120 frames per second. This quality of experience was only available to PC gamers who could invest thousands of dollars in dedicated gaming rigs.

Novel Gaming Experiences. The team also believed the cloud could unlock fundamentally new gaming experiences. This led to the creation of "Stadia Exclusive Features" (SEFs). One such exclusive feature, for example, was State Share. With State Share, a player could share a particular moment of their game with their friends. Their friends would then be able to hop into the game and start playing from that same moment.[9] SEFs opened up a new world of possibilities for gamers and game developers. The team hoped that these exclusive features would entice premium gamers to try the platform.

Choosing a business model

Zimring's founding vision for Stadia was to create a games subscription, similar to Netflix and Spotify, that would offer a diverse catalog of offerings across AAA and indie games. Almost instantly, AAA studios rejected the idea of selling their games through a third-party subscription. While subscriptions could broaden the reach of their games, they would also eat up much of the profits. Several entertainment executives were still traumatized by the havoc streaming services had wrought on the music industry. Before the development of digital music platforms, artists and labels had been making record profits selling $15-$20 CDs. The advent of file-sharing applications like Napster had left artists and labels with little choice but to join the streaming revolution, accepting fractions of a penny per stream. And unlike the music and movie industries, which saw thousands of new releases each year, there were only about 20-30 AAA games released each year. These AAA games drove consumer preference for various platforms. Furthermore, games were a "hits" driven business -- successful games could return huge multiples on their investment. For instance, Rockstar Games' *Grand Theft Auto V*, one of the best-selling titles of all times, cost $200M to develop but generated over $8B in revenues.[10] Studios had little incentive to give this up.

Thus, the Stadia team believed it had no choice but to settle for a traditional digital retail model, selling copies of games through its online store. Instead of selling a subscription for *games*, the team settled on selling a subscription to premium platform *features* called Stadia Pro. Pro cost $9.99 per month and allowed users to stream games in 4k resolution, gave them access to the best gaming servers, and more. For users who didn't want these premium features, Stadia Base would be available for free but would limit them to 1080p resolution. Both Base and Pro users would still have to purchase individual games from Stadia's online store, however.

Stadia's technical design

The aftermath of OnLive's demise had left the industry skeptical of the feasibility of cloud gaming. Zimring and his team would have to deliver an exceptional streaming experience to convince users of its merits, especially considering that Stadia's business model involved charging users for access to a premium streaming experience comparable to consoles.

Streaming games was much more challenging than streaming music or TV because it was *bidirectional*. Music was streamed one-way from a data center to the user's device. But games required input from users through a game controller. This input would have to be sent to the game server, processed, and the resulting frame would be sent back to the user's screen. This whole process would have to happen within 80 milliseconds, literally faster than the blink of an eye, in order to provide a smooth user experience.[11] Further, the system would have to deliver this demanding performance level continuously over the course of the entire gaming session. Any hiccups in this process would result in users experiencing lag, significantly diminishing the game experience. In order to pull this off, the Stadia team resolved that it needed to own every hardware and software component of the technology.

On the hardware side, the team decided that they would need to design custom, high-end gaming servers. The team enlisted AMD to design a purpose-built GPU for Stadia's servers instead of using off-the-shelf parts. But this came with a downside—the time and cost associated with designing and integrating custom parts into custom servers in Google's data centers -- all before a single user was on the platform. In a similar vein, the team also decided it needed to own the entire software stack. For example, the team decided to build a custom operating system for Stadia's servers instead of using Microsoft Windows, the industry standard. While this design would improve the streaming experience, using non-standard software made it more difficult for game developers to bring their games to Stadia. Games already developed to be run on Windows would require significant redevelopment to port over to the Stadia platform.

These decisions stood in sharp contrast to other cloud gaming efforts such as NVIDIA's GeForce Now service (GFN). GFN used standard NVIDIA GPUs and also ran Microsoft Windows. But Google's design choices appeared to have paid off when *Project Stream*, an early preview of Stadia, was met with an overwhelmingly positive response. Beta testers were shocked by the absence of lag and the smoothness of the gameplay. **Exhibit 6** shows a snapshot of discussion about the service from Reddit.[12]

Unveiling Stadia

At the annual Game Developers Conference in March 2019 in San Francisco, Zimring and his team unveiled Stadia to the world. Google CEO Sundar Pichai delivered the keynote address to open the conference, and he was followed by other executives, including Phil Harrison, a longtime Sony veteran and then General Manager of Stadia, and Majd Bakar, the Head of Engineering.

Harrison unveiled Stadia's ability to enable gameplay from any screen in a live demo. Bakar then shared information on Stadia's gaming servers. He revealed that Stadia's custom GPUs boasted 10.7 teraflops (TF) of performance, noting that this exceeded the capabilities of the Xbox One X (6.0 TF) and PlayStation 4 Pro (4.2 TF) combined (see **Exhibit 7**). **Exhibit 8** compares the technical specifications of Stadia and popular consoles. With such advanced servers, Bakar continued, Stadia could stream games in 4K resolution at 60 frames per second. He mused that in the future, Stadia would be able to stream games at 8K resolution and 120 frames per second.

Stadia also announced major partnerships with leading game studios such as Id and Ubisoft, signaling that major AAA titles were coming to the platform. But most notably, Google announced the launch of their own internal game studio, dubbed Stadia Games and Entertainment (SG&E). Led by industry veteran Jade Raymond, SG&E's charter was to develop compelling, first-party exclusive games for Stadia that took advantage of the platform's unique capabilities. To build out SG&E, Google had even acquired a small gaming studio called Typhoon Studios.

Stadia Launch

Stadia launched on November 19, 2019, in 14 countries around the world. Reviews from both fierce critics and smitten advocates came pouring in. Critics noted that while Stadia's streaming quality was impressive, the experience still didn't measure up to playing on a console or PC. Other critics were confused by the pricing model, which consisted of individual games and an optional subscription to Pro. The most universal criticism, however, was that Stadia had a mere 22 games to offer at launch, though the team announced that 120 more games were scheduled for release in 2020.[13]

But among the detractors, there were also users who had fallen in love with the platform. These advocates noted that the platform worked well enough for their needs and was cheaper and more convenient than traditional consoles. One Reddit forum called "StadiaDadia" gained over 2,000 members. The name played on the fact that several members of the forum, and several Stadia customers, were dads. Stadia perfectly fit their busy schedules and tight budgets.[14] **Exhibit 9** and **Exhibit 10** show a few posts from the forum.

Internally, Zimring and his team were disappointed with the launch. Based on *Project Stream's* reception, the team had been worried about a "success disaster" -- a situation where demand for the service outstripped the number of available game servers, forcing users to have to "wait in line" for a server. Fearing the impact this experience would have on Stadia's brand and on cloud gaming overall (the main value proposition was games "on-demand" after all), the team had built out excess data center capacity based on hefty user projections.

To the team's dismay, launch sales had come in far below forecast. Industry insiders estimated that Stadia had roughly 100,000 users at the end of 2019.[15] It was estimated that Stadia had invested over $1 billion in high-end gaming servers and that about half of this spend could be attributed to ongoing operating costs, such as data center space and power.

Rising pressure

The situation had hardly improved by August 2020. The team had achieved modest user growth through promotional campaigns, such as offering two months of Stadia Pro for free. But user numbers remained in the low hundreds of thousands, with most users on Stadia Base and a smaller proportion on Stadia Pro.

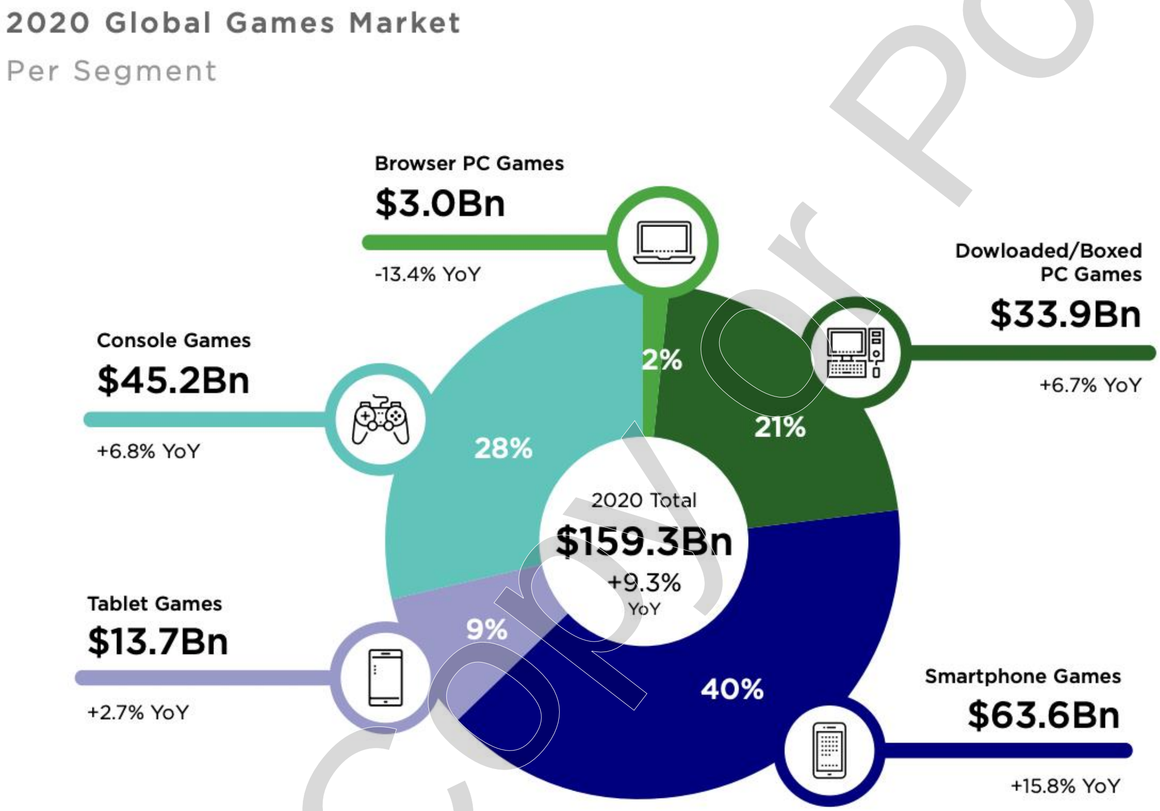
Adding to Zimring's worries was the fierce response from Microsoft and Sony. Both firms had unveiled their next generation consoles, the Xbox Series X and the PlayStation 5, which would be available starting November 2020. The new consoles boasted higher performance than Stadia's servers and offered premium features, such as support for 8k resolution and "ray-tracing," a rendering technique that resulted in more photorealistic graphics. Microsoft also announced its own cloud gaming service called XCloud.[16]

Acquiring more content to bolster Stadia's fledgling catalog was also proving to be a challenge. Some studio executives who had initially expressed interest in bringing their games to Stadia had a sudden change of heart and were now "pursuing other opportunities." This likely meant they had negotiated exclusive deals with Microsoft or Sony. The fact that Stadia wasn't running on Windows didn't help.

The decision

With mounting server and content costs threatening to overwhelm Stadia's fledgling business, Zimring needed to convince Google's senior leadership to continue investing in Stadia. For that, he would need a credible strategy and plan for growing the business. Many on Stadia's leadership team believed that their current plan, which was to focus on growing Pro subscriptions, was the right one. To achieve this, Stadia needed to bring more hit AAA games to the platform. It also needed to invest in new, upgraded servers to compete with Microsoft and Sony's next generation consoles. This would require a big check from Google's coffers, but also gave Stadia a shot at taking a meaningful bite out of the $80 billion console market. At the same time, Zimring considered communities like "StadiaDadia," whose members didn't need or want premium features like 4k streaming. He also thought about the rise of F2P games and the opportunities they presented. Perhaps instead of pursuing traditional AAA games, Stadia could grow the audience for F2P games and monetize through ads or in-game purchases. Zimring needed to decide on a strategy to bring to Google's CEO. The window to save Stadia was closing quickly.

Exhibit 1 2020 Global Games Market



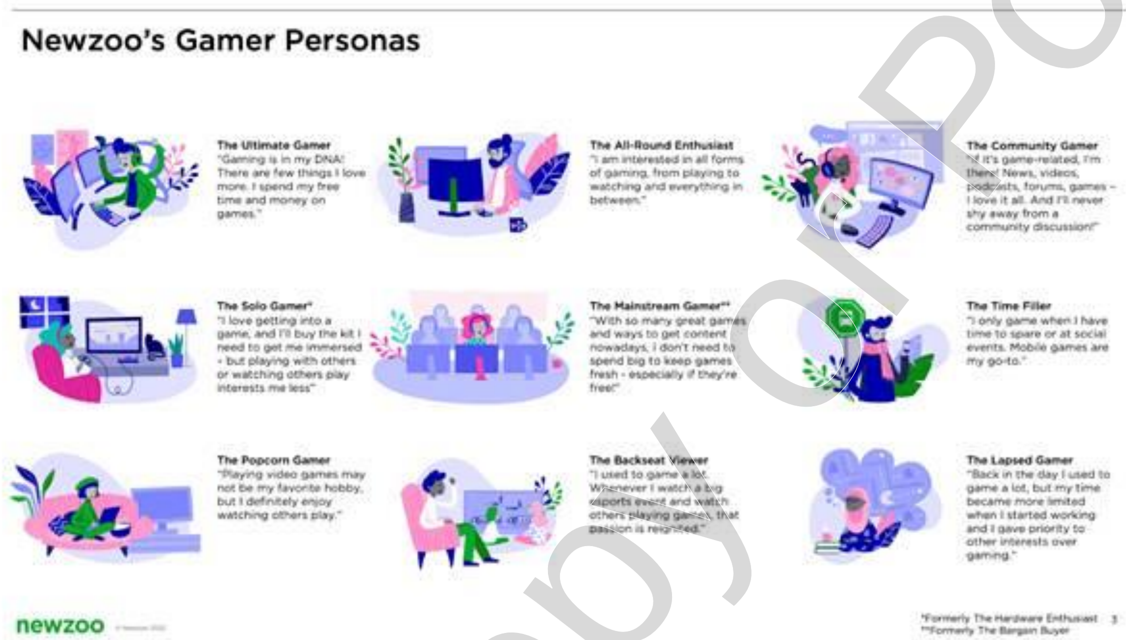
Source: "2020 Global Games Market Report," *Newzoo International BV*, 2020, p. 15 https://resources.newzoo.com/hubfs/Reports/2020_Free_Global_Games_Market_Report.pdf, accessed March 2024.

Exhibit 2 Recommended GPU for Fortnite vs Cyberpunk 2077

	Fortnite	Cyberpunk 2077
Business Model	Free to play	Buy to play
Recommended GPU	NVIDIA GTX 660 2.0 TFLOPS 140 Watts Price at launch: $229	NVIDIA GTX 2060 7.2 TFLOPS 175 Watts Price at launch: $399

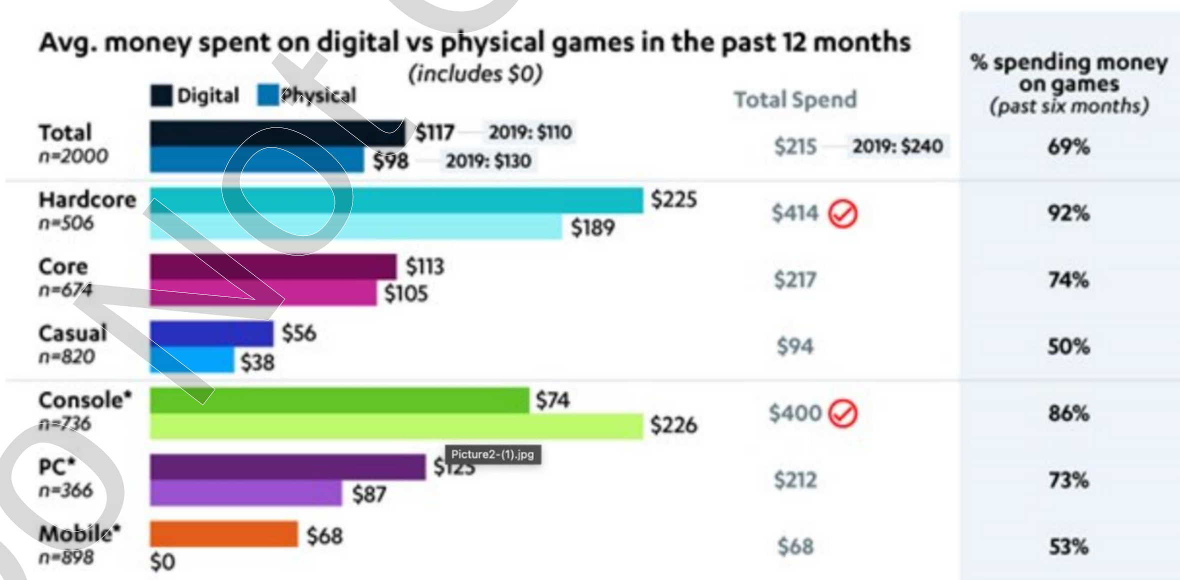
Source Casewriter's diagram based on JT Hussey, "Fortnite System Requirements," System Requirements Lab, June 18, 2020, https://www.systemrequirementslab.com/cyri/requirements/fortnite/15349, accessed March 2024, and "CyberPunk 2077 System Requirements," System Requirements Lab, https://www.systemrequirementslab.com/cyri/requirements/cyberpunk-2077/13169, accessed March 2024.

Exhibit 3 Newzoo's Gamer Personas



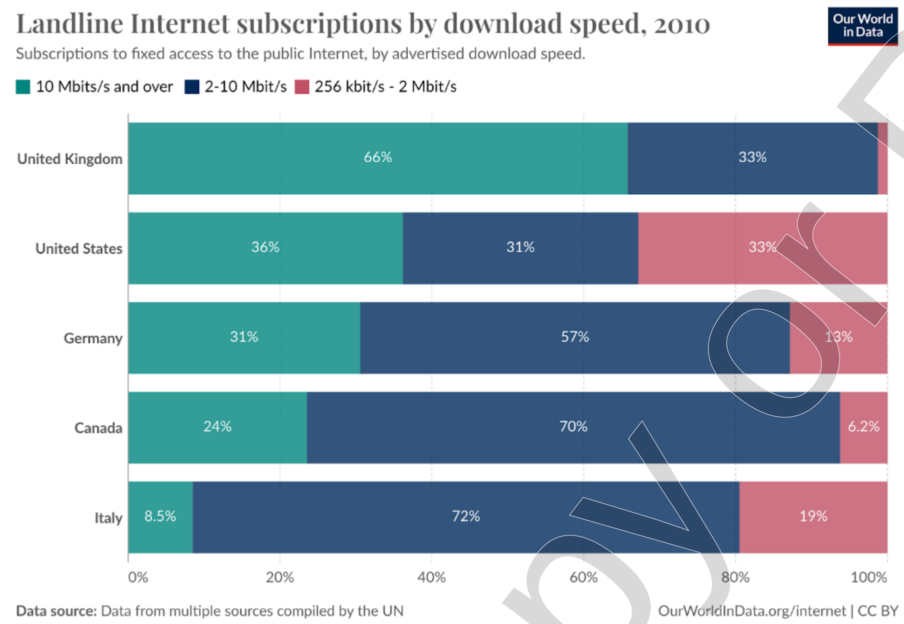
Source: "Newzoo's Gamer Segmentation," *Newzoo International BV*, 2022, https://core-rpg.net/Files/2022_Newzoo_Gamer_Segmentation_Full_Overview.pdf, accessed March 2024.

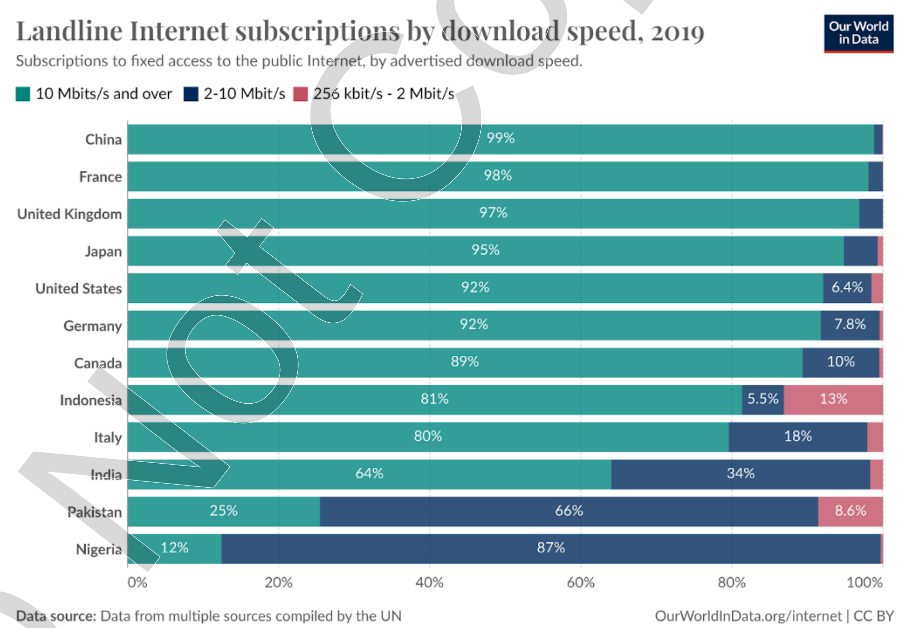
Exhibit 4 Average spend by gamer segment



Source: Jim Fellinger, "Immersive, Accessible Experiences Drive Growth in U.S. Gaming Market," Consumer Technology Association Market Research, September 26, 2022, https://www.cta.tech/Resources/Newsroom/Media-Releases/2022/September/2022-CTA-Gaming-Study, accessed March 2024.

Exhibit 5 Landline Internet Subscriptions by Download Speed, 2010 vs 2019





Source: Hannah Ritchie, Edouard Mathieu, Max Roser, and Esteban Ortiz-Ospina, "Internet," *Our World in Data*, 2023, https://ourworldindata.org/internet, accessed March 2024. Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0).

Exhibit 6 Review of Project Stream, Reddit

Ricochet888 · 5y ago

It ran extremely well. The input delay was pretty much non-existent. I'm sure in a multiplayer game, there would be some issues with input delay, but I didn't notice any in Project Stream.

I never measured the FPS but it was very smooth, with no noticeable framerate dips. I'd prefer to play single player games on Stadia. You don't have to worry about what specs your machine has, a shitty $300 Chromebook will run it just as well as a high end gaming PC, the games start instantly, it was just a matter of clicking Play and the game would boot in a second.

17 Reply Share ···

2 more replies

hongkongjack · 5y ago

I was really impressed by it. I tried to play it over ethernet as much as possible, but even on wifi there were plenty of times I forgot I wasn't playing on a console. The one thing you notice is the quality of the textures, which is why I'm excited that they were able to optimize the streaming to do 4K.

10 Reply Share ···

1 more reply

randomdharmabum · 5y ago

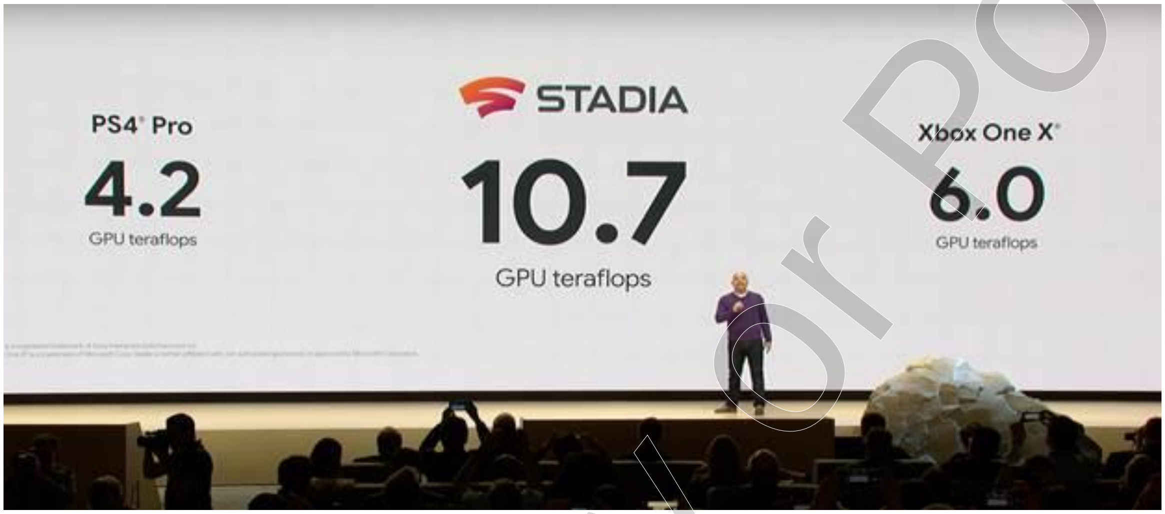
It was pretty damn nice. 1080p @60fps mostly, only every once in awhile did the resolution and fps drop and i would get lag, but we are talking about for a few seconds and then everything would be back to beautifully smooth. Never experienced noticeable input lag during gameplay, and I put over 100 hours into the game. I was playing mostly on a 2012 MacBook Air. I also played a bit of the game on my buddy's Chromebook, which we casted to his TV, just to see how it worked on a different network and machine. Casting was laggy, but that's expected, the game played beautifully on the chromebook. I especially liked that you are able to pick up exactly where you left off regardless of the computer you are on. We tried to measure the amount of data we were using, but couldn't get a read out on his network.

If Stadia works as well as Project Stream, I would be happy to pay for it even with a few blips here and there. Honestly i think i had more blips in the last game i played on my gaming rig than with Project Stream. Hope this helps.

6 Reply Share ···

Source: "R/ProjectStream on Reddit: Review of ProjectStream before Stadia?" *Reddit*, March 22, 2019, https://www.reddit.com/r/ProjectStream/comments/b4441v/review_of_projectstream_before_stadia/, accessed March 2024.

Exhibit 7 Stadia at Game Developer's Conference



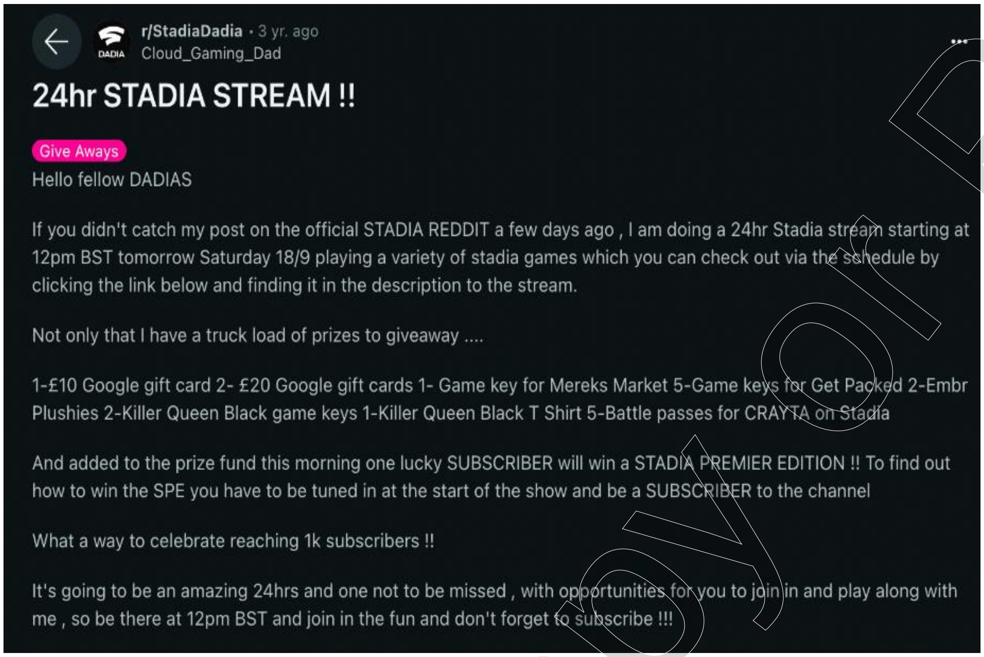
Source: IGN, "Full Google Stadia Connect Full Conference – E3 2019," YouTube, published June 6, 2019, https://www.youtube.com/watch?v=8igzue_fNDQ&t=1s, accessed March 2024.

Exhibit 8 Stadia vs Consoles Technical Comparison

	PS4 Pro	Nintendo Switch	Xbox One X	Stadia	Xbox Series X	PlayStation 5
Released on	Nov, 2016	Nov, 2017	Nov, 2017	Nov, 2019	Nov, 2020	Nov, 2020
CPU architecture	AMD Jaguar	NVIDIA Tegra	Custom AMD CPU	Custom Intel CPU	AMD Zen2	AMD Zen2
# of CPU cores	8	4	8	8	8	8
GPU architecture	AMD RDNA	NVIDIA Tegra	AMD RDNA	Custom AMD GPU	AMD RDNA 2.0	AMD RDNA 2.0
GPU TFLOPS	4.2	1.0	6.0	10.7	12.2	10.3
Cost at launch	$499	$299	$399	Stadia Pro: $9.99/mo Base: Free	$499	$499

Source: Casewriter's diagram based on Brendan Graeber et al., "Google Stadia vs. PS4 Pro vs. Xbox One X Comparison Chart," *IGN*, June 6, 2019, https://www.ign.com/wikis/stadia-google-game-console/Google_Stadia_vs._PS4_Pro_vs._Xbox_One_X_Comparison_Chart, accessed March 2024; Brendan Graeber et al., "Nintendo Switch Hardware Specs," *IGN*, July 10, 2019, https://www.ign.com/wikis/nintendo-switch/Nintendo_Switch_Hardware_Specs, accessed March 2024; and "Xbox Series X," *Microsoft*, https://www.xbox.com/en-US/consoles/xbox-series-x, accessed March 2024.

Exhibit 9 r/StadiaDadia, Reddit



r/StadiaDadia • 3 yr. ago
Cloud_Gaming_Dad

24hr STADIA STREAM !!

Give Aways

Hello fellow DADIAS

If you didn't catch my post on the official STADIA REDDIT a few days ago , I am doing a 24hr Stadia stream starting at 12pm BST tomorrow Saturday 18/9 playing a variety of stadia games which you can check out via the schedule by clicking the link below and finding it in the description to the stream.

Not only that I have a truck load of prizes to giveaway ....

1-£10 Google gift card 2- £20 Google gift cards 1- Game key for Mereks Market 5-Game keys for Get Packed 2-Embr Plushies 2-Killer Queen Black game keys 1-Killer Queen Black T Shirt 5-Battle passes for CRAYTA on Stadia

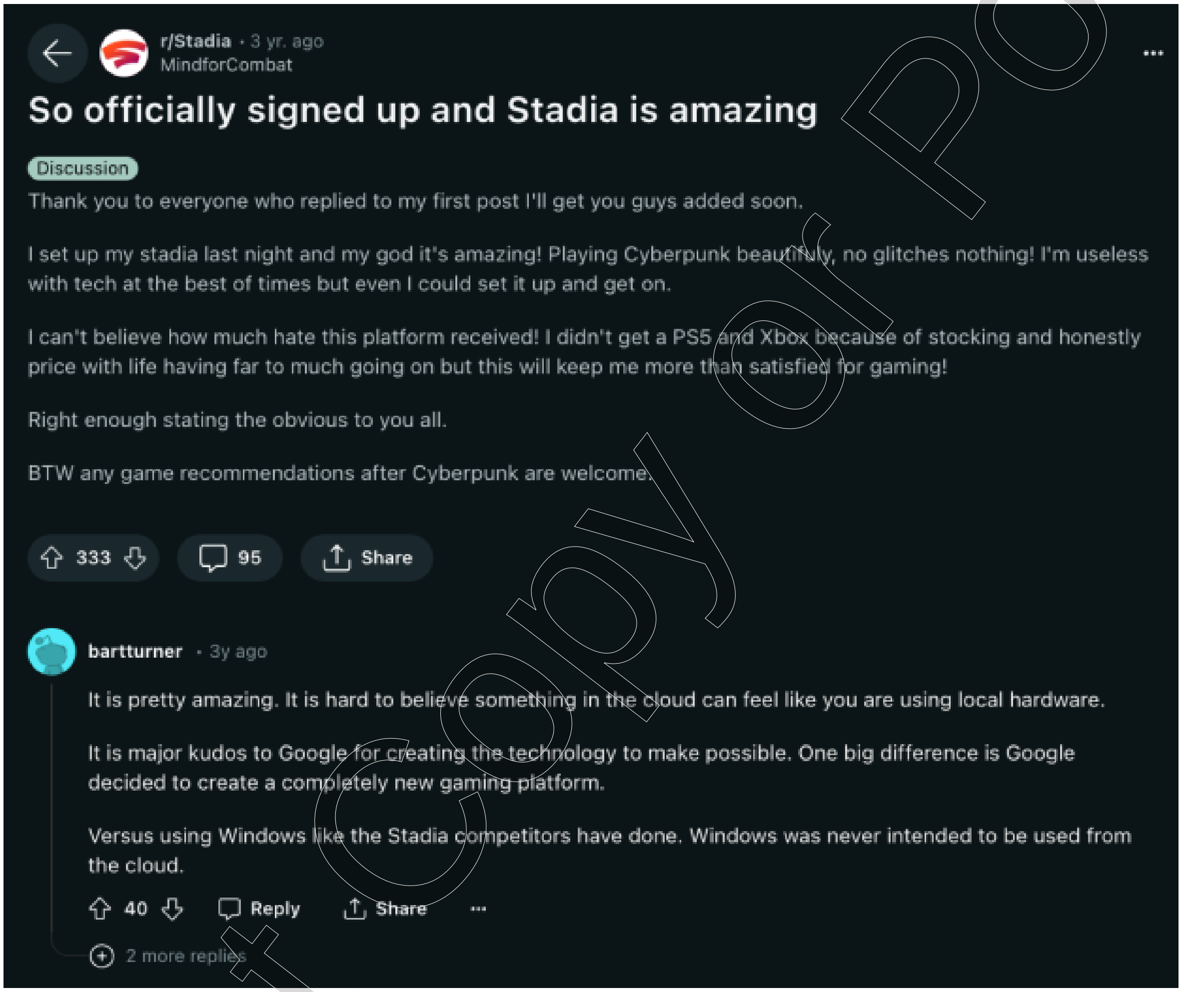
And added to the prize fund this morning one lucky SUBSCRIBER will win a STADIA PREMIER EDITION !! To find out how to win the SPE you have to be tuned in at the start of the show and be a SUBSCRIBER to the channel

What a way to celebrate reaching 1k subscribers !!

It's going to be an amazing 24hrs and one not to be missed , with opportunities for you to join in and play along with me , so be there at 12pm BST and join in the fun and don't forget to subscribe !!!

Source: "Stadia Dadia," *Reddit*, https://www.reddit.com/r/StadiaDadia/, accessed March 2024.

Exhibit 10 Reddit


r/Stadia • 3 yr. ago
MindforCombat

So officially signed up and Stadia is amazing

Discussion

Thank you to everyone who replied to my first post I'll get you guys added soon.

I set up my stadia last night and my god it's amazing! Playing Cyberpunk beautifully, no glitches nothing! I'm useless with tech at the best of times but even I could set it up and get on.

I can't believe how much hate this platform received! I didn't get a PS5 and Xbox because of stocking and honestly price with life having far to much going on but this will keep me more than satisfied for gaming!

Right enough stating the obvious to you all.

BTW any game recommendations after Cyberpunk are welcome.

333 95 Share

bartturner • 3y ago

It is pretty amazing. It is hard to believe something in the cloud can feel like you are using local hardware.

It is major kudos to Google for creating the technology to make possible. One big difference is Google decided to create a completely new gaming platform.

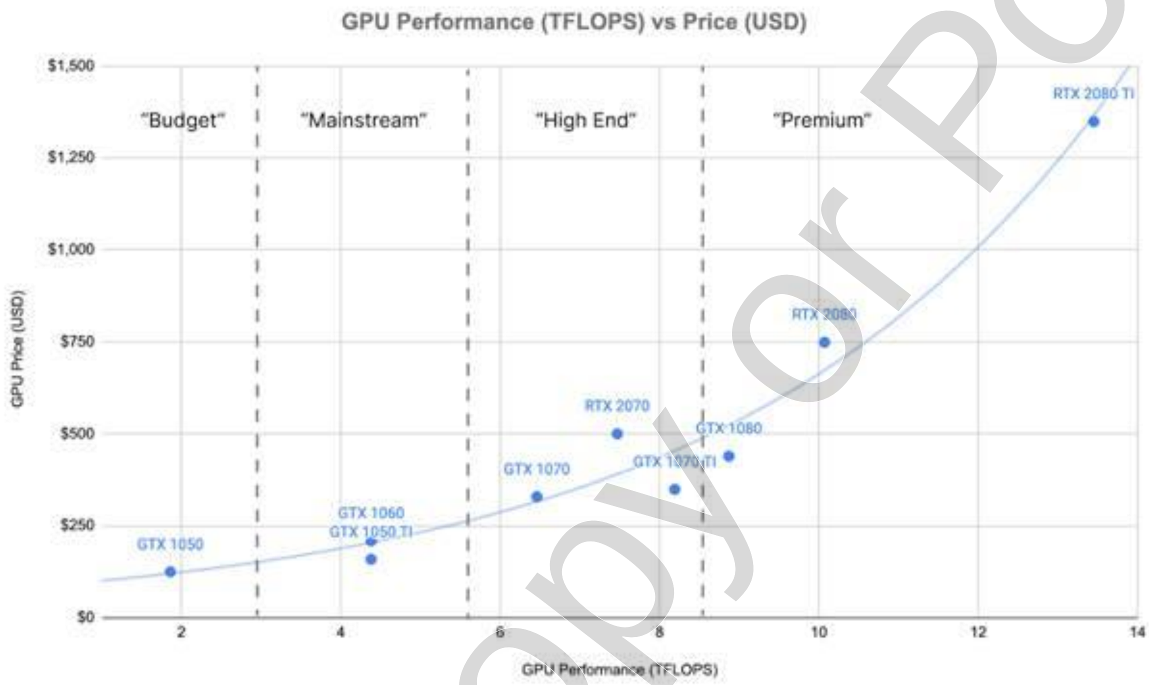
Versus using Windows like the Stadia competitors have done. Windows was never intended to be used from the cloud.

40 Reply Share

2 more replies

Source: "R/Stadia on Reddit: So officially signed up and Stadia is amazing" *Reddit*, January 23, 2021, https://www.reddit.com/r/Stadia/comments/l3a1ke/so_officially_signed_up_and_stadia_is_amazing/, accessed March 2024.

Exhibit 11 GPU Performance vs Price



Source: Casewriter's diagram based on Joel Hruska, "Best Graphics Cards for Every Budget," *ExtremeTech*, November 20, 2018, https://www.extremetech.com/gaming/273386-best-graphics-cards-for-every-budget, accessed March 2024.

Endnotes

[1] Gavin Divers, "Gaming Industry Dominates as the Highest-Grossing Entertainment Industry," Gamerhub, January 24, 2023, https://gamerhub.co.uk/gaming-industry-dominates-as-the-highest-grossing-entertainment-industry/, accessed March 2024.

[2] "Ready, Player Four Billion: The Rise of Video Games," *The Economist*, March 20, 2023, https://www-economist-com.ezp-prod1.hul.harvard.edu/special-report/2023/03/20/ready-player-four-billion-the-rise-of-video-games, accessed March 2024.

[3] "Game Industry Usage and Revenue Statistics 2023." HelpLama.com, January 8, 2024, https://helplama.com/game-industry-usage-revenue-statistics/, accessed March 2024.

[4] Alexander Smith, They Create Worlds: The Story of the People and Companies That Shaped the Video Game Industry Vol. 1: 1971 – 1982 (CRC Press, November 2019).

[5] Michael Cripe, "PlayStation Boss Jim Ryan Reveals PS2 Sold 160 Million Units Worldwide," IGN, March 29, 2024, https://www.ign.com/articles/playstation-boss-jim-ryan-reveals-ps2-sold-160-million-units-worldwide, accessed April 2024.

[6] Jeffrey Rousseau, "Ampere Analysis: Console Market Dipped to $56.2bn in 2022." GamesIndustry.biz, February 28, 2023, https://www.gamesindustry.biz/ampere-analysis-console-market-dipped-to-562bn-in-2022, accessed March 2024.

[7] Daniel Griffin, "The Era of Consolidation in the Games Industry," *Medium*, August 18, 2023, https://medium.com/@daniel.griffin/the-era-of-consolidation-in-the-games-industry-17727ce5d67#, accessed March 2024.

[8] Mark Hall and Willian L. Hosch, "Google," *Encyclopedia Brittanica*, February 28, 2024, https://www.britannica.com/money/Google-Inc, accessed March 2024.

[9] Jacob Creswell, "Google Stadia's State Share Is the Streaming Service's Most Ambitious Idea Yet," *CBR*, January 17, 2021, https://www.cbr.com/google-stadia-state-share/, accessed March 2024.

[10] Marcus Lu, "Charting Grand Theft Auto: GTA's Budget and Revenues," Edited by Pallavi Rao, *Visual Capitalist*, December 5, 2023, https://www.visualcapitalist.com/charted-grand-theft-auto-revenue-and-costs/#, accessed March 2024.

[11] Patricio Kobek, "Don't Expect To See Apex Legends (Or Other Stream-Intensive Games) On Stadia," *TheGamer*, June 17, 2019, https://www.thegamer.com/apex-legends-stadia-no/, accessed March 2024.

[12] "R/ProjectStream on Reddit: Review of ProjectStream before Stadia?" *Reddit*, March 22, 2019, https://www.reddit.com/r/ProjectStream/comments/b4441v/review_of_projectstream_before_stadia/, accessed April 2024.

[13] Sean Hollister, "Google Stadia Nearly Doubles Launch Lineup to 22 Games on Eve of Debut," *The Verge*, November 17, 2019, https://www.theverge.com/2019/11/17/20969937/google-stadia-cloud-gaming-launch-lineup-22-games-day-one, accessed March 2024.

[14] "Stadia Dadia," *Reddit*, https://www.reddit.com/r/StadiaDadia/, accessed March 2024.

[15] Dario Andric, "How Many People Use Google Stadia? – 2024 Statistics," *LEVVEL*, March 9, 2023, https://levvvel.com/google-stadia-statistics/, accessed March 2024.

[16] Will Tuttle, "Xbox is Building the Gaming Platform for the Next 20 Years," *Xbox Wire*, June 9, 2022, https://news.xbox.com/en-us/2022/06/09/whats-next-for-gaming/, accessed March 2024.